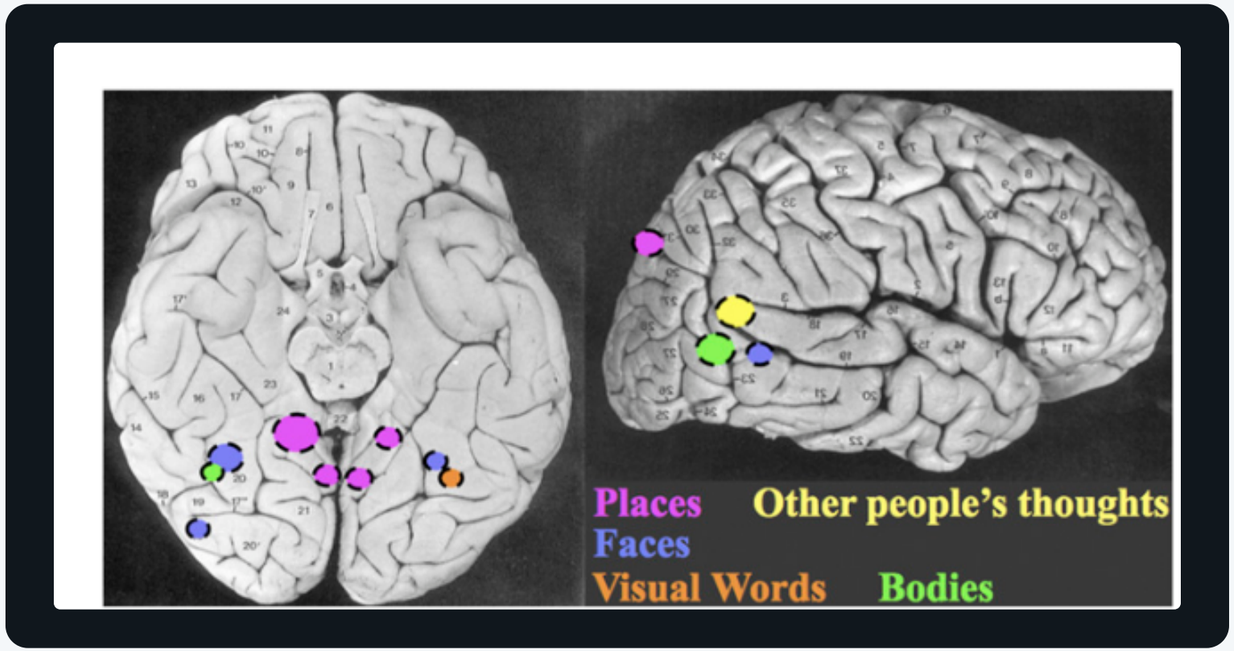
MIT 9.13 · LECTURE

Functional Specificity in the Human Brain

Architecture of the Mind

MIT 9.13 — Introduction to Cognitive Neuroscience
Prof. Nancy Kanwisher

● Faces ● Places ● Bodies ● Words ● Theory of mind



Functionally selective regions in the human brain: faces, places, bodies, words, and theory of mind.

Kanwisher 2010, p.2, Fig. 1

Agenda

Nine sections: from the historical question to the functional architecture of the mind.

01 Themes of the course

02 Why study the human brain?

03 The fundamental question
Functional localization

04 Methods
fMRI and ROI analysis

05 The face-selective region (FFA)

06 Other selective regions
PPA, EBA, rTPJ, VWFA

07 Implications
Developmental origins

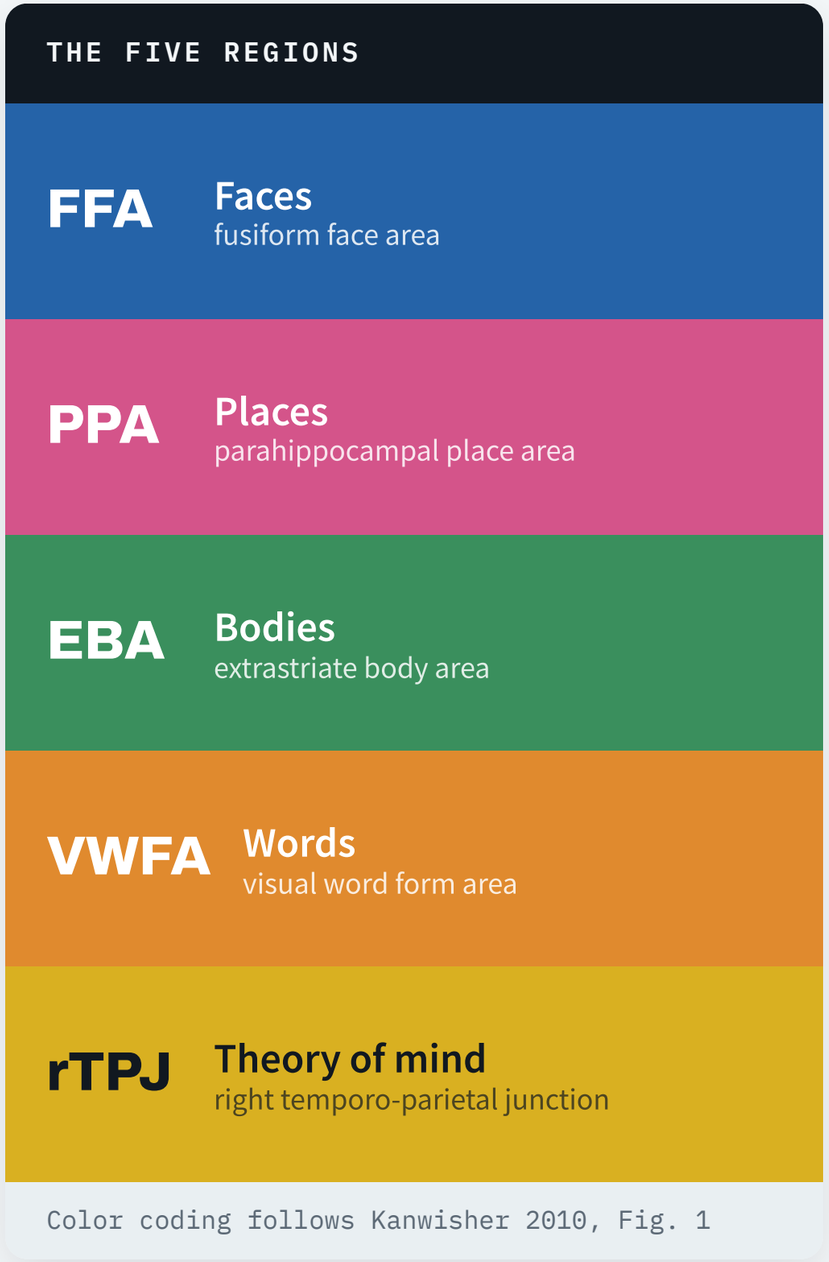
08 Summary
The functional architecture of the mind

09 References

Themes of the course

The human brain as a window into the functional architecture of the mind.

- **Functional specificity** — distinct regions for distinct cognitive functions
- **From debate to evidence** — historical controversy resolved by modern neuroimaging
- **Case studies** — five specialized regions, each with its own color identity below
- **Developmental origins** — genes vs. experience in shaping selectivity
- **Implications** — what specialized regions reveal about the mind



Why study the human brain?

A model trained on ImageNet fails on human-perceptual categories

IMAGENET CATEGORIES

71%

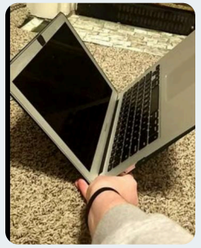
ResNet-200 accuracy, ImageNet-trained

↓ −46 pts

KATZ / BARBU CATEGORIES

25%

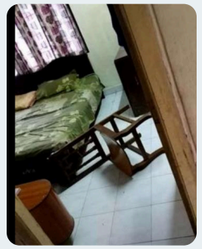
Same model, human-perceptual categories



KATZ / BARBU · OBJECT

Laptop

Object category example



KATZ / BARBU · PLACE

Bedroom

Place category example

> **Implication:** human-perceptual categories are not captured by ImageNet — the human brain organizes visual categories differently.

SECTION 2 SUMMARY

Themes: the brain as a window into the mind

1 The human brain is a window into the **functional architecture of the mind**.

2 Machine-learned categories (ImageNet) do not match human-perceptual categories (Katz / Barbu).

3 ResNet-200 accuracy drops from **71% to 25%** across this category shift.

4 This lecture examines the evidence for **functionally selective brain regions**.

material_1, p.1; Kanwisher 2010, Introduction

The fundamental question

THE CORE DEBATE

Is the human mind/brain composed of highly specialized components, each carrying out a specific aspect of cognition — or is it a general-purpose device in which each component participates in many processes?

TWO QUESTIONS STILL FIERCELY CONTESTED

1. How functionally specialized are brain regions?

   A matter of degree, not all-or-none.

2. Are only basic sensory and motor functions specialized — or also higher-level cognitive functions?

Kanwisher's argument: fMRI research is beginning to answer these questions with new clarity and precision.

Kanwisher 2010, Abstract and Introduction

Historical debate: localization vs. distribution

THE PROPOSAL THAT STARTED THE DEBATE

Franz Joseph Gall (1758–1828): the brain is the seat of the mind, and distinct mental faculties reside in specific brain organs.

In favor of localization

Distinct functions reside in distinct brain regions

Broca language localized to the left frontal lobe

Brodmann

Ferrier

Opposed to localization

Functions are distributed across the brain

Flourens

Golgi

Lashley arguments for distributed processing

BY THE EARLY 20TH C.

Consensus: at least basic sensory and motor functions reside in specialized brain regions. **The debate did not end — higher-level functions remain contested.**

Kanwisher 2010, Introduction, historical debate paragraph

SECTION 3 SUMMARY

The fundamental question

1. The central question: are cognitive functions implemented in **specialized components**, or in a **general-purpose device**?

2. Functional specialization is a **matter of degree**, not an all-or-none property.

3. Consensus exists for **basic sensory and motor localization**; higher-level functions remain contested.

4. **fMRI** provides the tools to resolve these questions with new precision.

Methods: fMRI and ROI analysis

STEP 1 · SIGNAL

fMRI principle

Blood flow increases **locally** in active regions of the brain — the basis of the fMRI signal.

Safe, noninvasive, millimeter-scale resolution in the normal human brain.

STEP 2 · COMPLEMENT

Lesion studies

Allow inferences about **necessity** — but clean functional dissociations are rare.

Lesions rarely affect all and only one region.

STEP 3 · ROI STRATEGY

Localize, then test

1

Localize

Functionally identify the region in each subject individually in a short "localizer" scan.

→

2

Test

Measure the response of that region in new conditions testing specific hypotheses.

Enables quantification across **dozens of stimulus and task manipulations**.

What makes a region “selective”?

Three criteria for strong evidence of functional specificity

01

CONSISTENT REPLICATION

Found in **dozens of studies across many labs**; present in virtually every neurologically intact subject.

02

LARGE EFFECT SIZE

Each region responds **about twice as strongly** to its preferred category as to any nonpreferred stimulus. Effect size determines the strength of the inference: knowing how to double a response gives a better handle on function than a small change.

≈2×

preferred vs. nonpreferred response

03

ROI QUANTIFICATION

Responses measured across **dozens of stimulus and task manipulations** in each subject.

Kanwisher 2010, §Neuroimaging Evidence, three reasons paragraph

SECTION 4 SUMMARY

Methods: how functional specificity is measured

1. **fMRI measures local blood flow increases** as an indicator of neural activity.

2. **Lesion studies establish necessity; fMRI establishes engagement** in normal brains. Both are needed to characterize a region's role.

3. **The ROI localize-and-test strategy** — find the functional "zip code" first, then test it — enables individual-subject quantification.

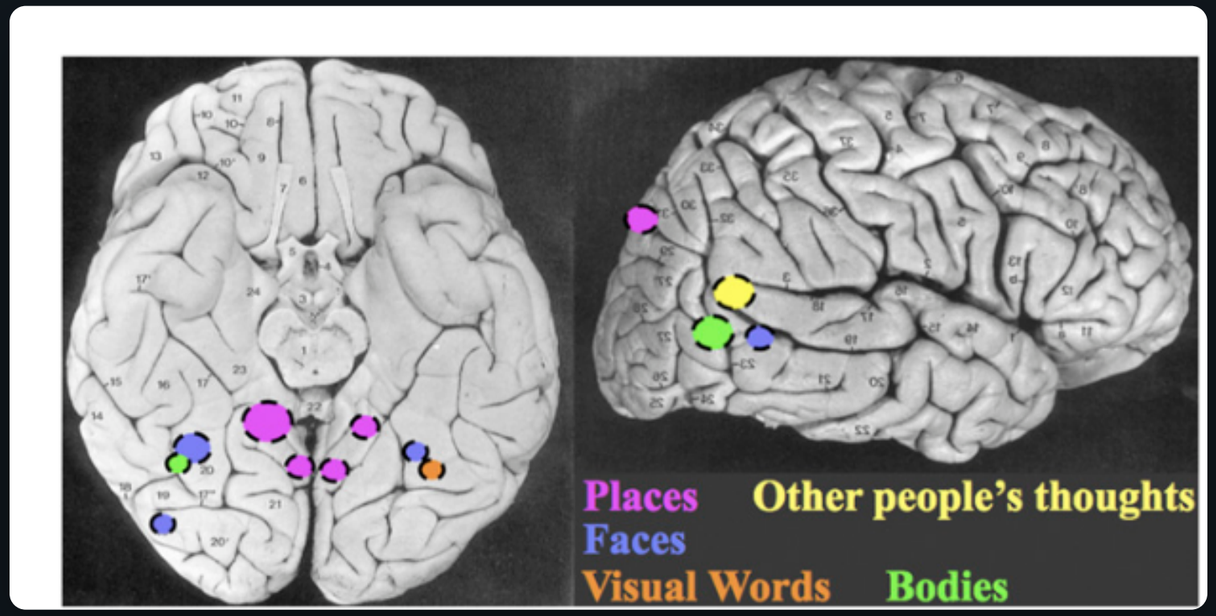
4. **Strong evidence for functional specificity** requires consistent replication, large effect sizes (≈2×), and extensive ROI quantification.

SECTION 5 · CASE STUDY

FFA: discovery and location

The fusiform face area (FFA) is one of the best-studied functionally selective regions in the human brain.

- Located in the **midfusiform gyrus** — on the bottom surface of the cerebral cortex, just above the cerebellum.
- Responds **significantly more strongly** when subjects view faces than objects.
- Found consistently in **dozens of studies across many labs**; present in virtually every neurologically intact subject.
- Part of the **basic functional architecture** of the human brain.



Approximate size and location of the major functionally selective regions. The **FFA (blue)** is highlighted in the legend. Kanwisher 2010, p.2, Fig. 1

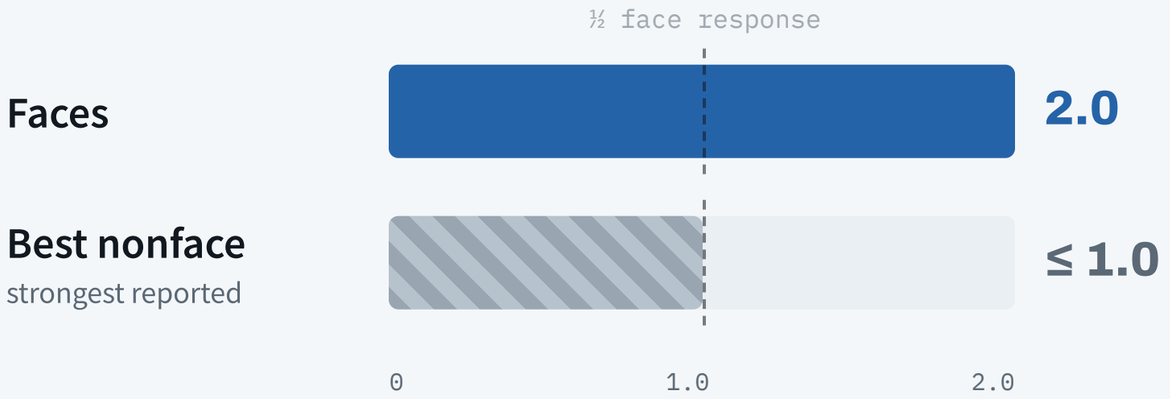
Kanwisher 2010, p.2, Fig. 1; §Neuroimaging Evidence, FFA paragraph

FFA: selectivity

The fusiform face area responds about twice as strongly to faces as to any nonface stimulus.

≈2×

Face response vs. the strongest nonface response, in the same region.



Relative FFA response (faces = 2.0). Conceptual illustration of reported effect sizes; not a plotted dataset.

> With high-resolution imaging (individual-subject analyses, no spatial smoothing), **no nonface object** has been reported to elicit more than **one-half** the face response.

RESPONDS SIMILARLY TO A WIDE VARIETY OF FACE IMAGES

- **Familiar faces**
- **Unfamiliar faces**
- **Schematic faces**
- **Cartoon faces**
- **Cat faces**
- Different sizes
- Different locations
- Different viewpoints

Kanwisher 2010, §Neuroimaging Evidence, FFA paragraph

FFA: robustness and behavioral signatures

BEHAVIORAL SIGNATURES MIRRORED IN THE FFA

DISPROPORTIONATE INVERSION EFFECTS

Face recognition degrades more when faces are inverted than when objects are inverted.

The FFA shows a neural correlate of this long-known behavioral effect.

HOLISTIC PROCESSING

The FFA is sensitive to holistic information in upright faces, but not in inverted faces.

The FFA is also sensitive to **face parts** (eyes, noses, mouths), the **T-shaped configuration** of features, and external features such as **hair**.

INVARIANCE AND IMAGINATION

INVARIANCE PROPERTIES

FFA representations show **some invariance** across changes in stimulus position, and **less invariance** across changes in viewpoint — mirroring behavioral results.

IMAGINATION

Imagining a face (eyes closed) selectively activates the **FFA**; imagining a place activates the **PPA**.

FFA: converging evidence from other methods

CONVERGING EVIDENCE

SUBDURAL ELECTRODES

Recordings from subjects undergoing presurgical mapping for epilepsy treatment show face-selective responses in approximately the FFA location.

LESIONS

Lesions in approximately the FFA location can produce selective deficits in face perception (prosopagnosia).

The FFA plays a central role in face perception but little if any role in the perception of nonface objects.

OPEN QUESTIONS

- ? What are the **precise representations and computations** performed by the FFA?
- ? Does the FFA play any **lesser role for nonfaces**?
- ? Is the FFA **cytoarchitectonically distinct**?
- ? What is its **connectivity**?
- ? Is it a **single contiguous region**?

Kanwisher 2010, §Neuroimaging Evidence, FFA paragraph (converging evidence and open questions)

SECTION 5 SUMMARY

The Fusiform Face Area (FFA)

1. The FFA in the **midfusiform gyrus** responds **about twice as strongly** to faces as to any nonface.

2. No nonface object exceeds **half the face response**; selectivity is robust across viewpoints, styles, and familiarity.

3. The FFA is the neural correlate of **inversion effects** and **holistic processing**; even imagining faces activates it.

4. Converging evidence from **subdural electrodes** and **lesion studies** supports its role.

5. The FFA is a **central component of the face-perception system** — some of the strongest evidence for functional specificity.

PPA: location and selectivity ● places

LOCATION

Defined functionally as the region **adjacent to the collateral sulcus** in parahippocampal cortex.

SELECTIVITY

Responds significantly more strongly to images of **scenes** than objects.

SCENE VARIETY

Responds to a wide variety of scenes:

`indoor` `outdoor` `familiar` `unfamiliar` `abstract (Legos)`

KEY FINDING · SPATIAL LAYOUT SUFFICIENCY

Floor + walls alone suffice.

Removing all objects from an indoor scene — leaving just the floor and walls — does **not** reduce the PPA response.

The PPA is primarily responsive to the spatial layout of one's surroundings, not to objects within scenes.

Reminiscent of the **geometric module** inferred from behavioral data in rats and human infants.

Kanwisher 2010, §Neuroimaging Evidence, PPA paragraph

PPA: necessity for navigation

NECESSITY EVIDENCE — LESION FINDINGS

Patients with damage in or near the PPA have difficulty encoding information about spatial layout — and more generally in knowing where they are.

This shows the PPA is not only **activated** when spatial layout is processed, but is **necessary** for this function.

The precise role of the PPA in place perception and navigation remains a topic of ongoing investigation.

RELATED REGIONS

RSC Retrosplenial cortex

Other scene-selective region; less well characterized than the PPA.

TOS Transverse occipital sulcus

Other scene-selective region; less well characterized than the PPA.

Kanwisher 2010, §Neuroimaging Evidence, PPA paragraph (necessity)

EBA: the extrastriate body area

LOCATION

Lateral surface of the brain, adjacent to — and sometimes partly overlapping — visual motion area **MT**.

SELECTIVITY

- Responds **significantly more strongly** to images of bodies and body parts than to objects or faces.
- Responds **equally** to visually very different body images: a photograph of a hand, a photograph of a body (human or animal), a schematic stick figure.

NECESSITY · LESION & TMS EVIDENCE

Disrupting the EBA impairs perception of body form — but not faces or object shape.

Brain lesion or transcranial magnetic stimulation (TMS) of the EBA selectively disrupts body-form perception, establishing the region as necessary for it.

WHAT THE EBA IS FOR

More engaged for other people's bodies than one's own.

More engaged for the **form / identity** of bodies than for the actions they carry out.

Kanwisher 2010, §Neuroimaging Evidence, EBA paragraph

rTPJ: theory of mind

Right temporo-parietal junction — discovered by Rebecca Saxe; identified in hundreds of subjects via the ROI method

Responds strongly

Scenarios describing what a person knows or thinks

- Knowledge and belief scenarios
  e.g., what another person knows or believes
- Deciding what another person knows
  task construed as pertaining to another's thoughts

Does not respond

Mental states without propositional content, or non-mental content

- Physical representations
  maps, photographs
- Physical appearance
  vivid descriptions without reference to the person's mind
- Bodily sensations
  thirst, hunger, pleasure

IDENTICAL STIMULI, IDENTICAL RESPONSE

rTPJ is **more strongly activated** when people decide what another person knows than when they make the identical response to the identical stimuli without construing the task as pertaining to another person's thoughts.

The most functionally selective high-level cortical region yet described in humans.

Kanwisher 2010, §Generality, rTPJ paragraph

VWFA: experience-derived selectivity

The visual word form area responds selectively to visually presented words — and its selectivity is shaped by reading experience.

≈1/10

VWFA volume relative to the FFA — so small it was not visible at standard imaging resolutions.

Schematic of the reported volume ratio — VWFA ≈ 1/10 of FFA



Responds severalfold higher to words than to line drawings of familiar objects. Consonant strings elicit the same response as words — meaning and orthographic regularity are not required.

Kanwisher 2010, §Origins, VWFA paragraph; Fig. S2 (SI Text)

SELECTIVITY FOLLOWS EXPERIENCE

UNFAMILIAR ORTHOGRAPHY

Low response to strings of digits and letters in Hebrew for English readers.

BILINGUALS (ENGLISH + HEBREW)

High response to words in both languages — the response is determined by the individual's experience.

CHINESE ILLITERATES

Character-selective response after several months of training, but not before.

SECTION 6 SUMMARY

Other selective regions: four more modules

1	PPA	Adjacent to the collateral sulcus; selective for **spatial layout**; necessary for navigation (lesion evidence).
2	EBA	Lateral surface near MT; selective for **bodies**; necessary for body form perception (TMS / lesion evidence).
3	rTPJ	Right temporo-parietal junction; selective for **thinking about others' thoughts**; the most selective high-level region.
4	VWFA	About **1/10 the volume of the FFA**; selectivity derived from experience, not genes.
5	Synthesis	Functional specificity extends from **basic perception to abstract cognition**.

Developmental origins: genes vs. experience

Both genes and experience play crucial roles in the development of all brain structures. The question: which details of each region's circuitry are specified in the genome, and which are derived from experience?

Genes

THE INNATE ACCOUNT

Rapid, accurate recognition of **faces, places, and bodies** had survival value.

Detailed wiring instructions for these circuits **may be specified in the genome**.

Experience

THE EXPERIENCE ACCOUNT

Each of us looks at **faces, places, and bodies** more frequently than almost any other stimulus class.

Cortical organization **can be affected by experience**.

CURRENT EVIDENCE

The face system may be **primarily genetically specified**, while the VWFA's selectivity is **primarily determined by experience**.

Kanwisher 2010, §Origins, introductory paragraph

Evidence for genetic specification of the face system

Five independent lines of evidence converge on a specific role of genes in face perception.

FAMILIES

Developmental prosopagnosia — a congenital disorder of face perception — runs in families.

TWINS

Face-perception ability is heritable: more strongly correlated for identical than fraternal twins, independent of domain-general abilities like IQ.

FMRI HERITABILITY

The spatial distribution of fMRI responses to faces across the ventral visual pathway is more similar between monozygotic than dizygotic twins — true for scenes, but not for chairs or words.

NEWBORNS

Impressive face discrimination abilities are present in human newborns.

MONKEYS

Baby monkeys reared for up to **2 years** without ever seeing faces still show face discrimination abilities.

CONCLUSION

These findings support the hypothesis that the **instructions for constructing the critical circuits for face perception are in the genome**.

Kanwisher 2010, §Origins, genes in face perception paragraph

Experience fine-tunes the face system

The basic face-perception system arises with virtually no face experience, then is fine-tuned by experience — paralleling language development.

First 3 days

Signatures at birth

Every aspect of face-specific processing tested so far — inversion effects, holistic processing — is present at the earliest ages ever tested; several signatures within the **first 3 days of life**.

Few months

Perceptual narrowing

Face discrimination, initially effective on **all races or primate species**, becomes restricted to only the race/species the subject has experienced.

Age 5–6

Adult-sized FFA

Adult-sized FFAs are found in the majority of children scanned at age 5 and 6 years.

Adulthood

Other race effect

We are better able to distinguish individuals from a **more familiar than a less familiar race**.

TAKEAWAY Genes provide the basic face system; experience narrows and fine-tunes it.

Kanwisher 2010, §Origins, experience and development paragraphs

SECTION 7 SUMMARY

Developmental origins: genes and experience

01 **The face system shows strong evidence for genetic specification.**
Heritability of face-selective responses; face discrimination in newborns and baby monkeys.

02 **Experience fine-tunes the face system.**
Perceptual narrowing within months of life; the other race effect in adulthood.

03 **The VWFA derives its selectivity from experience, not genes.**
Literacy training creates word-selective responses in previously illiterate readers.

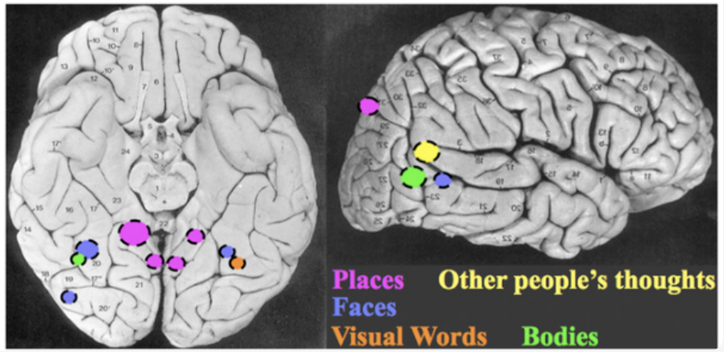
04 **The precise roles of genes and experience remain an active area of research.**
Different regions can have different origins: the face system is largely genetic, the VWFA is experience-derived.

The functional architecture of the mind

> **A cognitively precise parts list for the human brain is within reach.**

- **Strong evidence** for cortical regions selectively engaged in perceiving faces (FFA), places (PPA), bodies (EBA), and words (VWFA) — and in thinking about what other people are thinking (rTPJ).
- Possible specializations for **number, music, and language** are under active investigation.
- The most important question is not **where** each component is found, but **which functions get their own brain region** — and why some do and others do not.
- The cognitive functions implemented in specialized brain regions are strong candidates for **fundamental components of the human mind**.

RECAP · THE COLOR-CODED MAP



Kanwisher 2010, p.2, Fig. 1 — faces (blue), places (pink), bodies (green), words (orange), theory of mind (yellow).

Kanwisher 2010, §Conclusions; p.2, Fig. 1

Key takeaways

1. The human brain contains **functionally specialized regions** for faces, places, bodies, words, and theory of mind
2. Strong evidence requires consistent replication, large effect sizes ($\approx 2\times$), and extensive ROI quantification
3. Functional specificity extends from basic perception to **abstract cognition** (rTPJ)
4. The face system is largely **genetically specified**; the VWFA is **experience-derived**
5. A cognitively precise **parts list** for the human brain is within reach
6. The functions implemented in specialized regions are strong candidates for **fundamental components of the mind**